PETAL MEDICAL KNOWLEDGE BASE & CLINICAL GROUND TRUTH SPECIFICATION

Document Version: 1.1.0
Target Architecture: Petal Mobile Application (MVP - Flo Architectural Model)
Clinical Standards Reference:

- World Health Organization (WHO) *Medical Eligibility Criteria for Contraceptive Use* (5th Edition)
- WHO *Selected Practice Recommendations for Contraceptive Use* (3rd Edition)
- International Federation of Gynecology and Obstetrics (FIGO) Menstrual Disorders Working Group
- American College of Obstetricians and Gynecologists (ACOG) Clinical Guidelines
- DKT International / DKT Philippines *Trust Pill* Official Product Insert & Prescribing Information

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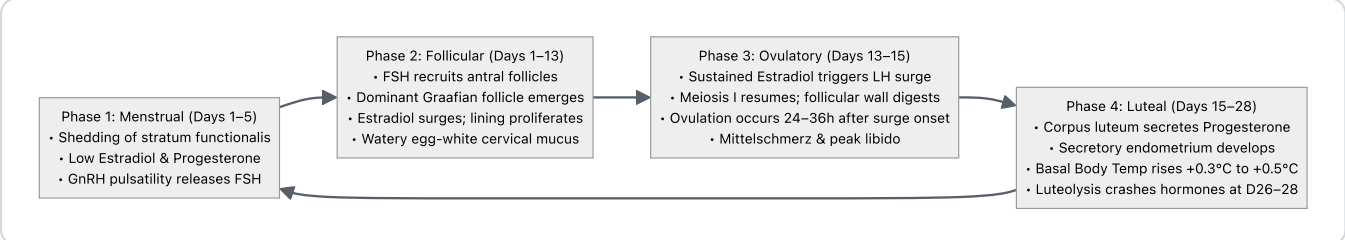
1. Natural Menstrual Cycle Mechanics & Flo-Style Tracking Logic

1.1 Clinical Definitions and Normal Population Baselines

In clinical gynecology and digital health algorithm design, a natural menstrual cycle is a bio-feedback endocrine loop governed by the Hypothalamic-Pituitary-Ovarian (HPO) axis.

- **Cycle Day 1 (CD 1):** The first day of full, bright red menstrual flow requiring sanitary protection. Premenstrual spotting (light brownish discharge) is biologically and algorithmically excluded from Day 1 and is classified as the terminal day(s) of the preceding cycle.
- **Normal Cycle Length Range: 24 to 38 days** (FIGO 2018 guidelines). A population reference average of **28 days** is used as an uncalibrated cold-start baseline.
- **Normal Menstruation Duration: 3 to 8 consecutive days** (average: 4 to 5 days).
- **Normal Menstrual Blood Loss: 5 to 80 mL** per cycle.
- **Normal Cycle-to-Cycle Variability:** Variations of ≤ 7 to 9 days between the shortest and longest cycles across a 12-month period are considered clinically regular in mature adult females.

1.2 Endocrinology and Biological Events Across the 4 Cycle Phases



Phase 1: Menstrual Phase (Cycle Days 1–5)

- **Endocrine Profile:** Estradiol (< 50 pg/mL) and Progesterone (< 1 ng/mL) are at nadir levels. The absence of corpus luteum steroidogenesis removes negative feedback on the hypothalamus, permitting pulsatile Gonadotropin-Releasing Hormone (GnRH) release, which triggers an early subtle rise in Follicle-Stimulating Hormone (FSH).

- **Endometrial Histology:** Arteriolar constriction and ischemia in the *stratum functionalis*, followed by tissue necrosis, proteolytic degradation, and shedding with blood and fluid.
- **Clinical Presentation & Symptoms:**
  - Uterine cramping/dysmenorrhea mediated by locally released Prostaglandin $F_{2\alpha}$ ($PGF_{2\alpha}$).
  - Systemic symptoms: Lethargy, low back pain, pelvic heaviness, low basal body temperature (BBT: $36.1^{\circ}\text{C} - 36.4^{\circ}\text{C}$).

Phase 2: Follicular / Proliferative Phase (Cycle Days 1–13)

- **Endocrine Profile:** Rising pituitary FSH recruits a cohort of 5–15 antral follicles. By Cycle Days 5–7, the follicle with the highest density of FSH receptors and aromatase activity emerges as the **dominant (Graafian) follicle**. Granulosa cells of this follicle synthesize escalating concentrations of 17-β-estradiol (E_2).
- **Endometrial Histology:** High circulating estrogen induces rapid mitotic proliferation of the *stratum functionalis*, elongating straight glands and growing spiral arterioles (endometrium thickens from $\approx 1\ \text{mm}$ to $8 - 12\ \text{mm}$).
- **Cervical Mucus Dynamics:** Estrogen induces mucin production by cervical crypt cells, shifting mucus from scant/thick to clear, thin, profuse, watery, and highly elastic (**spinnbarkeit** $\geq 8\ \text{cm}$), resembling raw egg white. This facilitates sperm capacitation and migration.
- **Clinical Presentation:** Rising physical energy, elevated mood, increasing sexual desire, clear skin.

Phase 3: Ovulatory Phase (Cycle Days 13–15 in a 28-Day Model)

- **Endocrine Profile:** When sustained estradiol concentrations surpass the clinical threshold of $\geq 200\ \text{pg/mL}$ for $\geq 48\ \text{hours}$, systemic feedback on the adenohypophysis switches from negative to **positive feedback**. This induces the catastrophic **Luteinizing Hormone (LH) surge** (accompanied by an FSH surge).
- **Follicular Dynamics:** The LH surge resumes oocyte meiosis I (arrested at prophase I) into metaphase II, stimulates local prostaglandins and proteolytic collagenases (plasmin, matrix metalloproteinases), weakens the follicular wall, and induces follicular rupture **24 to 36 hours after the LH surge onset** (or **10 to 12 hours after the peak LH level**).
- **Clinical Presentation:**
  - *Mittelschmerz* (mild unilateral lower abdominal twinge/cramping from follicular fluid or micro-hemoperitoneum).
  - Peak fertile egg-white cervical mucus (maximum Spinnbarkeit).
  - Occasional minor ovulatory spotting caused by a sudden temporary estrogen dip before progesterone catches up.
  - Peak female libido.

Phase 4: Luteal / Secretory Phase (Cycle Days 15–28)

- **Endocrine Profile:** The ruptured Graafian follicle collapses and luteinizes into the **corpus luteum**, vascularized by the theca cells and granulosa lutein cells. It synthesizes massive quantities of **Progesterone** (peaking at $10 - 25\ \text{ng/mL}$ around Cycle Day 21) alongside moderate Estradiol. High progesterone exerts intense negative feedback on LH and FSH, blocking secondary follicle maturation.
- **Endometrial Histology:** Progesterone converts the proliferative endometrium into a **secretory state**: uterine glands become tortuous and sacculated, secreting glycogen-rich fluid into the uterine cavity to nourish a prospective blastocyst.
- **Thermoregulatory Shift:** Progesterone acts on the preoptic hypothalamic thermoregulatory center, inducing a sustained rise in **Basal Body Temperature (BBT)** of $+0.3^{\circ}\text{C}$ to $+0.5^{\circ}\text{C}$ (0.5°F to 1.0°F), confirming retrospective ovulation.

- **Luteolysis:** In the absence of embryonic Human Chorionic Gonadotropin (hCG), the corpus luteum undergoes programmed apoptosis (luteolysis) on Cycle Days 26–28, forming the fibrous *corpus albicans*. Progesterone and estrogen crash catastrophically, triggering spiral arteriolar spasm, endometrial ischemia, and Cycle Day 1 of the subsequent cycle.
- **Clinical Presentation:** Premenstrual Syndrome (PMS) spectrum—breast tenderness (mastalgia), peripheral fluid retention, bloating, headache, emotional lability/irritability, food cravings, and acne vulgaris.

1.3 Flo-Style Mathematical Tracking & Prediction Algorithms

Algorithm 1: Cycle Length Estimation & Next Period Onset

Natural menstrual tracking engines (such as Flo) separate cycle history into an initial population prior, which transitions to individual rolling weighted averages as data points accumulate.

$$\hat{L}_k = \begin{cases} 28 & \text{if } k = 0 \text{ (Cold start: No user history)} \\ \frac{1}{k}\sum_{i=1}^{k} L_i & \text{if } 1 \le k < 3 \text{ (Arithmetic mean)} \\ \sum_{i=1}^{N} w_i \cdot L_{k-N+i} & \text{if } k \ge 3 \text{ (Rolling weighted historical average)} \end{cases}$$

Where:

- L_i = Length in days of recorded cycle i ($\text{Start Date}_{i+1} - \text{Start Date}_i$).
- N = Rolling window size (Standard Flo engine uses $N = 3$ to 6 completed cycles).
- w_i = Exponential or linear weighting factors favoring recent cycles, normalized such that $\sum_{i=1}^{N} w_i = 1.0$. (For a 3-cycle linear weight: $w_1 = 0.167, w_2 = 0.333, w_3 = 0.500$).

Next Predicted Period Start Date:

$$\text{Date}_{\text{NextPeriod}} = \text{Date}_{\text{CurrentPeriodStart}} + \hat{L} \text{ days}$$

Algorithm 2: Estimated Ovulation Day Calculation

In reproductive biology, the follicular phase is highly variable (ranging from 10 to 24+ days), whereas the **luteal phase duration (λ) is physiologically constant** at 14 ± 1.5 days across healthy reproductive populations.

Therefore, ovulation estimation is strictly computed **backwards** from the predicted next period onset date, never forward from the start of the previous period:

$$\text{Date}_{\text{Ovulation}} = \text{Date}_{\text{NextPeriod}} - 14 \text{ days}$$

$$\text{Cycle Day of Ovulation } (\text{CD}_O) = \hat{L} - 14 + 1$$

Example: For a woman with an average cycle length $\hat{L} = 32$ days:

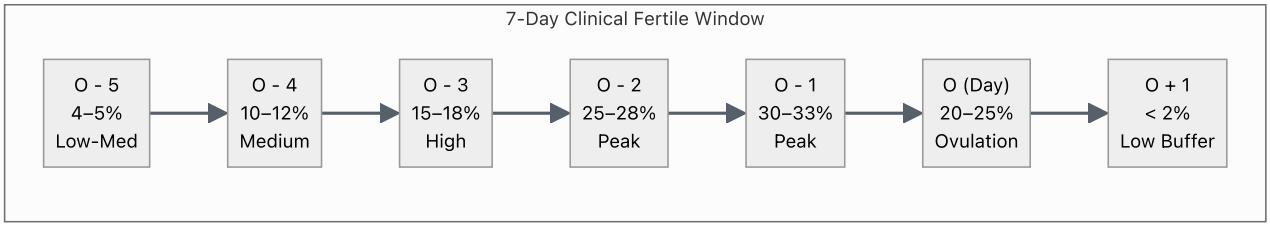
$$\text{CD}_O = 32 - 14 + 1 = \text{Cycle Day } 19$$

Algorithm 3: Fertile Window Duration & Conception Probability Gradient

The fertile window is bounded by the biological viability of gametes inside the female reproductive tract:

- **Sperm Viability in Fertile Cervical Mucus:** Up to 120 hours **(5 days)**.
- **Ovum Post-Ovulatory Lifespan:** 12 to 24 hours **(1 day)**.
- **Biological Fertile Window:** Exactly **6 days** ($O - 5$ through O).
- **App UI Display Window (Flo Standard Conservative Buffer):** Exactly **7 days**, incorporating a $+1$ day post-ovulation margin of error ($O + 1$) to accommodate luteal phase timing deviations ($\lambda \in [13, 15]$) and ovum survival.

$$\text{Fertile Window Interval} = [\text{Date}_{\text{Ovulation}} - 5 \text{ days}, \quad \text{Date}_{\text{Ovulation}} + 1 \text{ day}]$$



1.4 Algorithmic Handling of Cycle Irregularity & Outliers

Step 1: Outlier Detection and Filtering

Cycles meeting any of the following clinical outlier criteria are excluded from moving average calculations ($\hat{L}$) to prevent model poisoning:

1. **Extreme Absolute Length:** Cycle length $L_i < 21$ days or $L_i > 45$ days (unless validated by user confirmation of medical history or pregnancy loss).
2. **Statistical Outlier:** Length deviating from the user's running historical median by $> 3 \times \text{IQR}$ (Interquartile Range) or $> 2.5 \times \sigma$.

Step 2: Irregularity Index & Clinical Alert Flags

$$\Delta_{\text{Range}} = \max(L) - \min(L) \quad \text{across the past 6 to 12 cycles}$$

- **Regular Cycle:** $\Delta_{\text{Range}} \leq 7$ days. The app displays precise day-specific predictions.
- **Variable / Irregular Cycle:** $\Delta_{\text{Range}} \geq 8$ to 10 days, or Standard Deviation $\sigma > 4.5$ days.

App Logic Adjustment for Irregular Cycles:
The app transitions from deterministic single-day prediction points to a **Probabilistic Confidence Range**:

$$\text{Expected Onset Window} = [\text{Date}_{\text{CurrentPeriodStart}} + \min(L), \quad \text{Date}_{\text{CurrentPeriodStart}} + \max(L)]$$

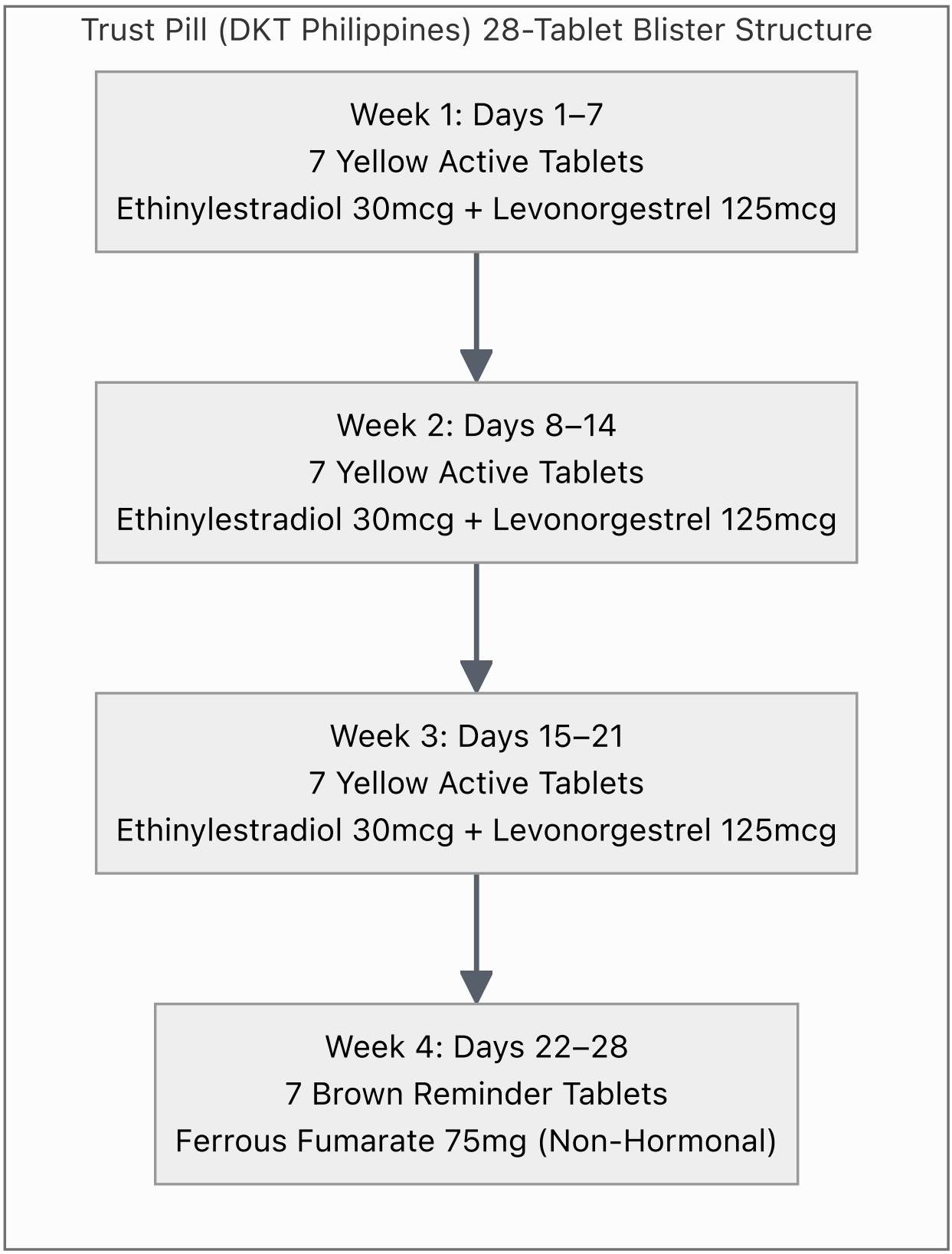
Fertile Window Widening:
When cycles are irregular, the fertile window expands across the potential ovulation range:

$$\text{Start} = \min(\hat{L}) - 14 - 5 \quad \text{to} \quad \text{End} = \max(\hat{L}) - 14 + 1$$

- **Clinical Warning Trigger (ACOG/FIGO Flag):**
  If consecutive cycle lengths are < 24 days (Polymenorrhea), > 38 days (Oligomenorrhea), or if bleeding is absent for > 90 days (Secondary Amenorrhea), the app triggers an algorithmic alert:
  "Your cycle shows significant variability outside the clinical baseline. Calendar predictions may be unreliable. We recommend consulting an OB-GYN to evaluate hormonal balance, thyroid function, or PCOS."

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2. Trust Pills (DKT Philippines) Product Specifications



2.1 Formulation, Chemistry, and Chemical Dosages

Trust Pill® (DKT Philippines Inc., FDA Registration No. DRP-336) is an oral, low-dose, monophasic combined oral contraceptive (COC) packaged in a 28-tablet blister card.

1. Active Tablets: 21 Yellow Round Tablets (Days 1–21)

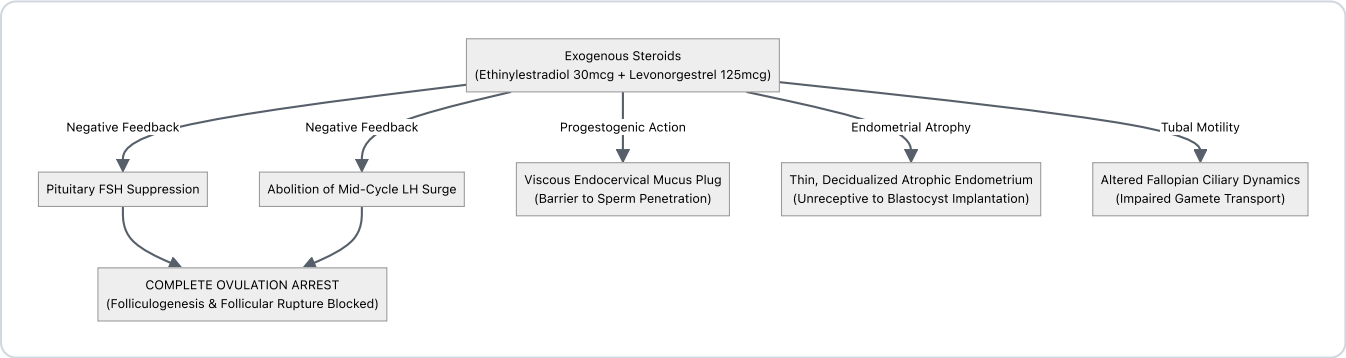
- **Ethinylestradiol (EE):** 30 mcg **(**0.030 mg**)**
  - *Class:* Synthetic 17-α-ethinyl-substituted steroidal estrogen.
  - *Molecular Formula:* $C_{20}H_{24}O_2$.
- **Levonorgestrel (LNG):** 125 mcg **(**0.125 mg**)**
  - *Class:* Second-generation synthetic 19-nortestosterone derivative (gonane family progestin).
  - *Molecular Formula:* $C_{21}H_{28}O_2$.
- *Excipients:* Lactose monohydrate, maize starch, povidone, magnesium stearate, yellow iron oxide / colorant.

2. Inactive / Reminder Tablets: 7 Brown Round Tablets (Days 22–28)

- **Ferrous Fumarate:** 75 mg
  - *Elemental Iron Content:* Approximately 24.75 mg of elemental iron (33% by weight).
  - *Hormone Content:* 0.00 mg **(Completely Non-Hormonal)**.
  - *Clinical Purpose:* Provides micro-nutritional replacement for iron lost during withdrawal bleeding to combat iron-deficiency anemia while enforcing the habit of taking one pill daily.

2.2 Mechanism of Action (Triple Redundant Barrier)

Combined oral contraceptives provide multi-level redundancy against conception:



1. **Primary Mechanism: Hypothalamic-Pituitary-Ovarian (HPO) Suppression**
   - Ethinylestradiol suppresses pituitary secretion of **FSH**, preventing antral follicle recruitment and follicular selection.
   - Levonorgestrel suppresses the hypothalamic GnRH pulse frequency and eliminates the mid-cycle **LH surge**. Without an LH spike, follicular rupture and ovulation are prevented in $> 99\%$ of compliant cycles.
2. **Secondary Mechanism: Cervical Mucus Hostility**
   - Levonorgestrel induces rapid progestogenic thickening of cervical mucus within 48 hours of administration. Mucus becomes scant, highly viscous, and cellular, reducing sperm mobility and preventing penetration into the endocervical canal.
3. **Tertiary Mechanism: Endometrial Inhospitability & Tubal Motility**

- Constant exposure to progestin disrupts the normal cyclic proliferation of the endometrium, producing early pseudo-decidualization followed by glandular exhaustion and thinning (*endometrial atrophy*). Even if escape ovulation were to occur, the atrophic endometrium cannot support blastocyst implantation.
- Progestin alters fallopian tube contractility and cilia beat frequency, impairing gamete and zygote transport.

2.3 Critical App Tracking Logic: Natural Menses vs. Withdrawal Bleeding

The Core Physiological Divergence

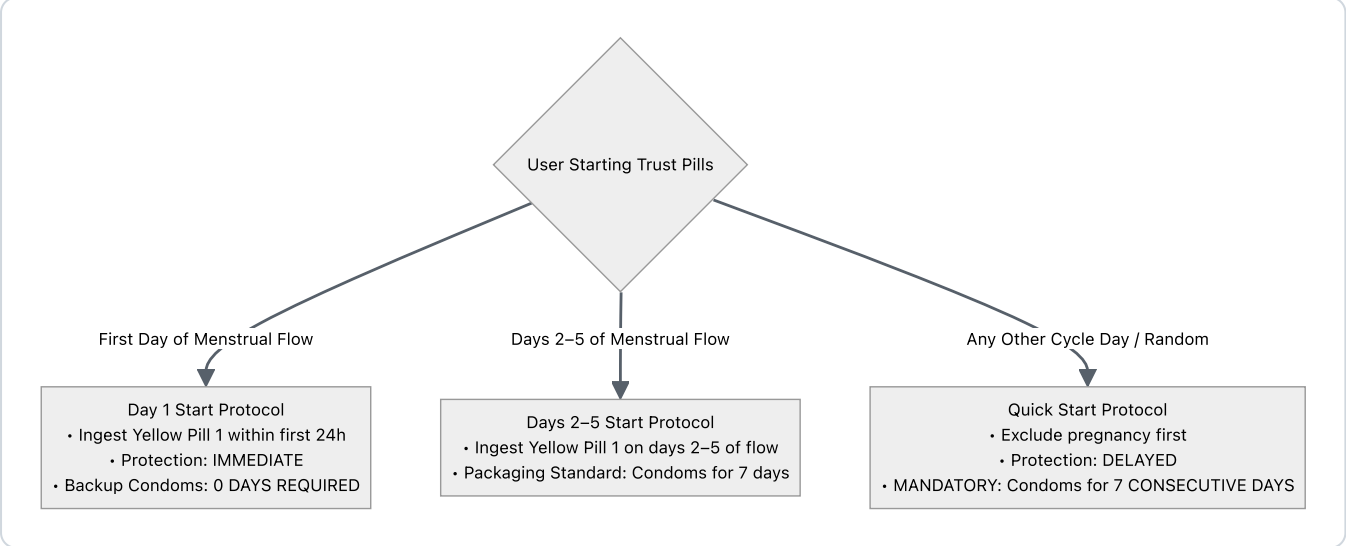
- **Natural Menstruation:** The culmination of a completed ovulatory cycle. The corpus luteum dies, endogenous progesterone plunges, and a fully developed secretory endometrium sheds.
- **Withdrawal Bleeding:** A pharmacologically engineered bleed occurring during Days 22–28 (the 7 brown iron pill days). It is caused by the sudden drop in exogenous hormones after stopping the yellow active pills. Bleeding on Trust Pills is typically lighter, shorter (2–4 days), and less painful than natural menses because the endometrium remains thin and atrophic.

Algorithmic Directive for Petal Mobile App

> **IMPORTANT:** **MANDATORY APP RULE: SUPPRESS NATURAL CYCLE ALGORITHMS ON ACTIVE PILLS** When a user selects **"Trust Pills (28-day COC)"** in Petal: 1. **DISABLE ALL OVULATION PREDICTIONS:** Do NOT display "Estimated Ovulation Day", "LH Peak", or "Conception Chance". Displaying an ovulation date or fertile window to a user correctly taking Trust Pills is medically false and can cause unnecessary anxiety or misuse of calendar-based contraception. 2. **CONVERT CALENDAR UI TO PACK CYCLE:** Replace the 28-day natural menstrual wheel with a **28-Day Pack Progression Bar** (Days 1–21 Active Phase → Days 22–28 Hormone-Free Interval / Withdrawal Bleeding Phase). 3. **EXPECTED BLEEDING WINDOW:** Predict withdrawal bleeding to initiate on **Pack Days 23–25** (typically 2 to 3 days after taking the last active yellow pill, during the brown iron tablets).

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3. Administration, Routine, and Pack Transitions



3.1 Pack Initiation Protocols

Protocol A: Day 1 Start (Gold Standard / Manufacturer Recommended)

- **Procedure:** User consumes Yellow Pill 1 on the **first 24 hours of natural menstrual bleeding** (Cycle Day 1).
- **Contraceptive Protection: Immediate**. Follicular recruitment is arrested before it begins.
- **Backup Barrier Method (Condoms): 0 Days Required**.

Protocol B: Days 2–5 Start

- **Procedure:** User consumes Yellow Pill 1 between Day 2 and Day 5 of the onset of natural menses.
- **Clinical Safety Rule:** While WHO guidelines note that protection is largely established if started within 5 days of menses onset, **standard clinical practice and DKT safety guidelines mandate using condoms for the first 7 consecutive days of active pills** to account for variability in early follicular recruitment.

Protocol C: Quick Start (Any Day of Cycle / Outside Menses)

- **Procedure:** User starts Yellow Pill 1 on any day of the month, provided pregnancy is reasonably excluded (e.g., negative high-sensitivity urine hCG test and no unprotected intercourse within the past 14 days).
- **Contraceptive Protection:** Delayed.
- **Backup Barrier Method: Mandatory for 7 consecutive days** of active yellow pill consumption.

Protocol D: Switching from Other Contraceptive Methods

- **From Another 28-Day COC:** Take Yellow Pill 1 of Trust Pill the day after the last active hormonal pill of the previous pack, or at the latest, the day after the last inactive tablet. No backup needed.
- **From a Progestin-Only Pill (POP / Minipill):** Can switch on any day. Must use backup condoms for the **first 7 consecutive days**.
- **From an Injectable (DMPA / Depo-Provera):** Start Trust Pill on the day the next scheduled injection is due. No backup needed if started on time.
- **Post-Abortion (First Trimester):** Start on Day 1 post-procedure; protection is immediate.

3.2 Daily Dosing Schedules & The "Grace Window"

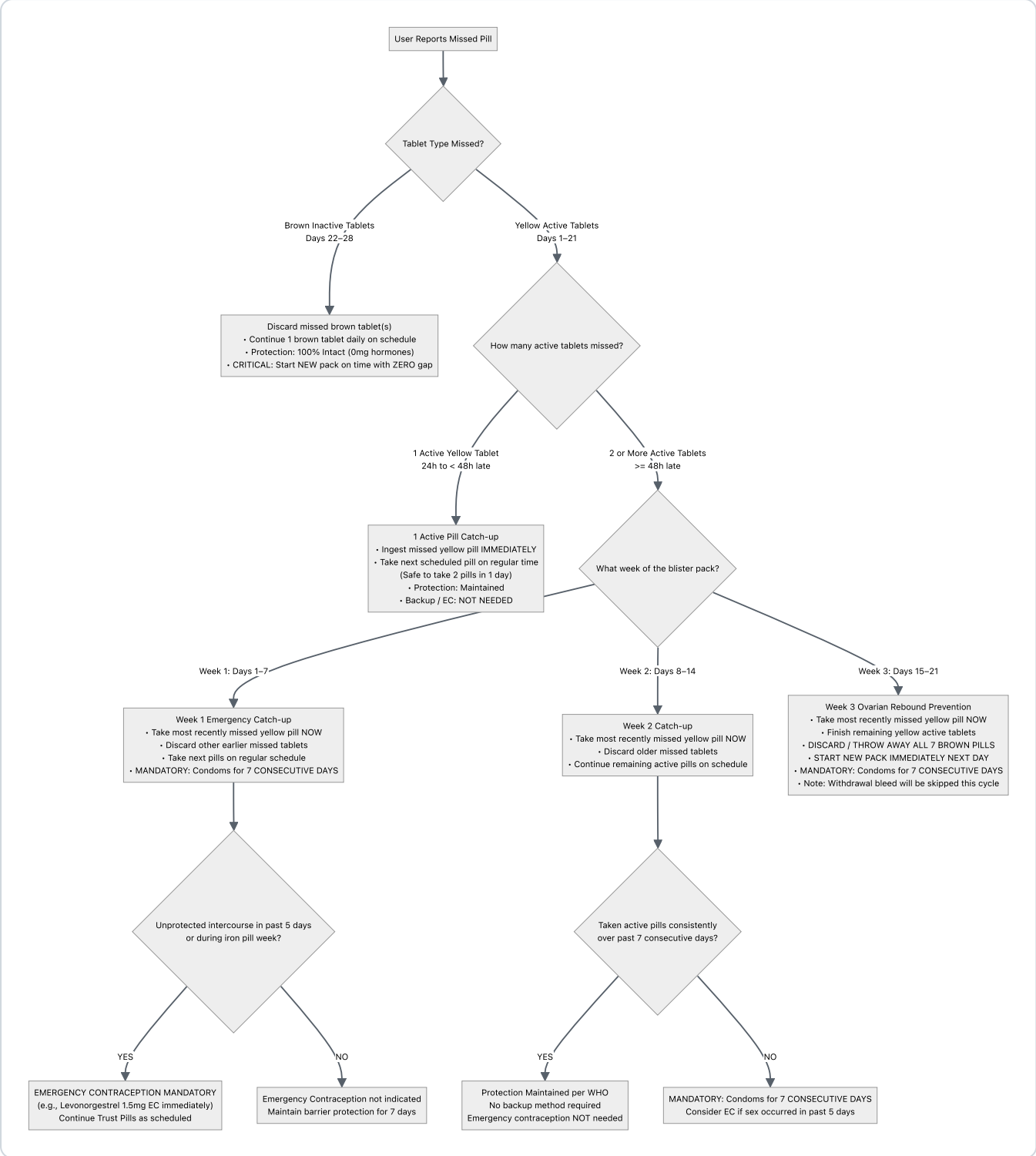
- **Daily Dosing Interval:** Exactly **one tablet every 24 hours**.
- **Anchoring Habit:** Advise users to anchor pill intake to a fixed daily event (e.g., after dinner, brushing teeth at night, or a 9:00 PM alarm). Taking the pill with an evening meal or before bed reduces potential estrogen-induced nausea.
- **Pharmacokinetic Tolerance Window ("Grace Period"):**
  - Because Trust Pills contain Ethinylestradiol (30 mcg) and Levonorgestrel (125 mcg), the plasma elimination half-life ($t_{1/2}$) of levonorgestrel is prolonged (≈ 24 to 32 hours), and ethinylestradiol is ≈ 18 to 26 hours.
  - **On-Time:** Taken within **0 to 12 hours** of the designated scheduled daily time.
  - **Late:** Taken between **12 and 24 hours** late. Contraceptive efficacy remains intact.
  - **Missed:** A delay of ≥ 24 **hours** past the scheduled time (or when two scheduled pills are due at the same time).

3.3 Seamless Pack Transitions (The "Zero-Gap" Rule)

- **28-Day Unbroken Continuity:** The Trust Pill regimen is a **continuous cycle**.
- **Day 28 Transition Rule:** When the user consumes Brown Pill 7 (Tablet 28), they **MUST open and start Tablet 1 (Yellow Pill 1) of a new pack the very next day**.
- **Do NOT Wait for Bleeding to End:** Users must never wait for withdrawal bleeding to finish before opening the next pack. Withdrawal bleeding may continue into the first few yellow pills of the new pack, or it may conclude early.
- **Zero Pill-Free Interval:** There must be **ZERO days without a pill**. Any hormone-free gap exceeding 7 days allows pituitary FSH levels to rebound, leading to rapid follicular recruitment, escape ovulation, and method failure.

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4. Missed Pill Decision Trees & Algorithmic Logic



4.1 Definitive Classification (WHO SPRC & CDC Standards)

- **Late Pill:** Taken < 24 hours after the regular scheduled time (< 48 hours elapsed since the last pill was taken).
- **Missed Pill (1 Tablet):** Taken 24 to < 48 hours after the regular scheduled time (48 to < 72 hours elapsed since the last pill was taken).
- **Missed Pills (≥ 2 Consecutive Tablets):** ≥ 48 hours have elapsed since a tablet should have been ingested (≥ 72 hours elapsed since the last pill was swallowed).

4.2 Modular IF/THEN Decision Logic

Scenario 1: Missed EXACTLY ONE Active Yellow Pill (Days 1–21)

- **Time Delta:** ≥ 24 hours and < 48 hours past scheduled dosing time.

```
IF (missed_active_pills == 1) THEN
    Action_Immediate: "Take the missed yellow pill immediately right now."
    Action_Routine:   "Take your next yellow pill at your regular scheduled time."
    NOTE:             "This may mean taking 2 pills in one day or at the exact same moment. Thi
    Backup_Protection:"No backup barrier method (condom) is required."
    EC_Indication:    "Emergency contraception is NOT needed."
    Efficacy_Status:  "Contraceptive protection is maintained."
ENDIF
```

Scenario 2: Missed TWO OR MORE Active Yellow Pills in WEEK 1 (Pack Days 1–7)

- **Time Delta:** ≥ 48 hours late during Days 1–7.
- **Clinical Vulnerability: CRITICAL DANGER ZONE**. The 7-day hormone-free interval (from the previous pack's brown pills) has been inadvertently extended. This can trigger follicular recruitment and escape ovulation.

```
IF (missed_active_pills >= 2 AND pack_day >= 1 AND pack_day <= 7) THEN
    Action_Immediate: "Take the most recently missed active yellow pill immediately right now."
    Action_Discard:   "Discard / leave the other missed active tablets in the blister pack."
    Action_Routine:   "Continue taking your remaining active pills daily at your usual time."
    Backup_Protection:"MANDATORY: Use condoms or abstain from sex for the NEXT 7 CONSECUTIVE DA

    IF (unprotected_sex_occurred_in_past_5_days == TRUE OR unprotected_sex_during_iron_week ==
        EC_Indication: "TAKE EMERGENCY CONTRACEPTION (EC) IMMEDIATELY."
        EC_Details:    "Levonorgestrel 1.5mg EC is recommended. Continue your Trust Pills as di
    ELSE
        EC_Indication: "Emergency contraception not required if no sex occurred in the past 5 d
    ENDIF
ENDIF
```

Scenario 3: Missed TWO OR MORE Active Yellow Pills in WEEK 2 (Pack Days 8–14)

- **Time Delta:** ≥ 48 hours late during Days 8–14.
- **Clinical Vulnerability:** Moderate. Prior 7 days of active pills successfully suppressed the HPO axis.

```
IF (missed_active_pills >= 2 AND pack_day >= 8 AND pack_day <= 14) THEN
    Action_Immediate: "Take the most recently missed active yellow pill immediately right now."
    Action_Discard:   "Discard / leave any other earlier missed active pills."
    Action_Routine:   "Continue taking remaining active yellow pills daily at your regular time

    IF (active_pills_taken_consistently_past_7_days == TRUE) THEN
        Backup_Protection: "No backup method required per WHO guidelines."
        EC_Indication:     "Emergency contraception is NOT needed."
    ELSE
        Backup_Protection: "MANDATORY: Use condoms for the next 7 CONSECUTIVE DAYS."
        EC_Indication:     "Consider EC if unprotected sex occurred in the last 5 days."
    ENDIF
ENDIF
```

Scenario 4: Missed TWO OR MORE Active Yellow Pills in WEEK 3 (Pack Days 15–21)

- **Time Delta:** ≥ 48 hours late during Days 15–21.
- **Clinical Vulnerability: HIGH OVARIAN REBOUND RISK**. Missing pills here, followed by the planned 7-day brown pill interval, creates an extended low-hormone window that can lead to ovulation.

```
IF (missed_active_pills >= 2 AND pack_day >= 15 AND pack_day <= 21) THEN
    Action_Immediate: "Take the most recently missed active yellow pill immediately right now."
    Action_Discard:   "Discard any other missed yellow pills."
    Action_Routine:   "Continue taking active yellow pills daily until all 21 yellow pills are

    CRITICAL_TRANSITION_RULE:
    "DO NOT TAKE THE 7 BROWN REMINDER PILLS!"
    "Throw away / discard all 7 brown pills from this pack completely."
    "Start Tablet 1 (Yellow Pill) of a BRAND NEW PACK the very next day after your 21st yellow

    Backup_Protection: "MANDATORY: Use condoms or abstain for the NEXT 7 CONSECUTIVE DAYS."
    User_Counseling:   "You will likely NOT experience withdrawal bleeding this month. Breakthr
    EC_Indication:     "Consider EC if unprotected sex occurred and pills were missed earlier i
ENDIF
```

Scenario 5: Missed Inactive Brown Reminder Pills (Pack Days 22–28)

- **Hormonal Impact:** None (0 mg hormones; contains only Ferrous Fumarate).

```
IF (missed_pills ARE brown_reminder_pills) THEN
    Action_Discard:   "Throw away the missed brown tablet(s) from the blister card."
    Action_Routine:   "Continue taking the remaining brown tablets, one per day, at your usual
    CRITICAL_RULE:    "Start your NEW PACK on the scheduled day. DO NOT DELAY the start of the
    Backup_Protection:"NO backup method needed. Contraceptive protection is 100% unaffected."
    EC_Indication:    "Emergency contraception is NOT needed."
ENDIF
```

4.3 Gastrointestinal Upset Protocols (Vomiting & Diarrhea)

Active hormones are primarily absorbed in the stomach and upper duodenum within **2 hours** of oral intake.

Acute Emesis (Vomiting within 2 to 4 Hours of Ingestion)

```
IF (vomiting_occurred_within_4_hours_of_active_yellow_pill == TRUE) THEN
    Clinical_Status: "The tablet was not fully absorbed. This is medically equivalent to a MISS
    Action:          "Take another active yellow pill from a separate backup blister pack IMMED
    Next_Step:       "Take your next regular pill tomorrow at your usual scheduled time."
    Backup_Pills:    "If no spare pack is available, take the next yellow pill from the current
ENDIF
```

Severe Watery Diarrhea (> 24 to 48 Hours)

```
IF (severe_watery_diarrhea_persists_over_24_hours == TRUE) THEN
    Clinical_Status: "Intestinal transit is accelerated, impairing steroid absorption and enter
    Action:          "Continue taking your daily Trust Pills as scheduled."
    Backup_Rule:     "MANDATORY: Use condoms during the diarrheal illness AND for 7 CONSECUTIVE
ENDIF
```

5. Side Effects, Warning Signs & Contraindications

5.1 Expected Adaptation Side Effects (Months 1–3)

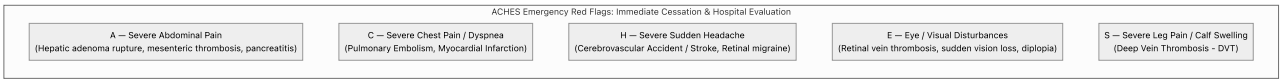
During the initial 90-day acclimatization window, the exogenous hormones stabilize the endometrium and downregulate the endogenous HPO axis. Users should be reassured that these symptoms are temporary and typically resolve without medical intervention.

Side Effect	Underlying Mechanism	Actionable Mitigation Strategy
Breakthrough Bleeding / Spotting	Fragile, transitioning endometrium; micro-vessel shed	Take pill at the **EXACT same hour**; do NOT stop pill; reassure user
Nausea / Dyspepsia	Estrogen stimulation of gastric mucosa and central CTZ receptor	Take pill with evening meal or immediately before bedtime
Breast Tenderness (Mastalgia)	Estrogen fluid retention & ductal epithelial stimulation	Wear supportive non-underwire bra; reduce dietary sodium
Mild Headache	Fluctuating systemic estrogen levels during early cycles	OTC analgesics (Paracetamol); maintain consistent hydration

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5.2 Critical Emergency Red Flags: The "ACHES" Acronym

The **ACHES** acronym identifies severe, life-threatening complications—specifically Venous Thromboembolism (VTE), Deep Vein Thrombosis (DVT), Pulmonary Embolism (PE), Myocardial Infarction (MI), and Ischemic Stroke.

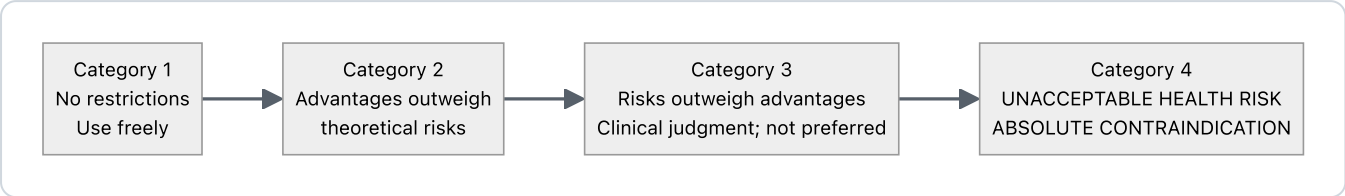


> **CAUTION:** **IF A USER LOGS ANY "ACHES" SYMPTOM: STOP PILLS IMMEDIATELY AND DIRECT TO EMERGENCY CARE**

- **A — Abdominal Pain (Severe / Unremitting):** Hepatic adenoma rupture, mesenteric vein thrombosis, acute pancreatitis, gallbladder disease.
- **C — Chest Pain (Sharp / Radiating / Shortness of Breath / Hemoptysis):** Pulmonary Embolism (PE), acute coronary syndrome / myocardial infarction.
- **H — Headaches (Severe, Sudden Onset / Migraine with Aura / Neurological Deficits):** Cerebrovascular accident (stroke), sagittal sinus thrombosis, malignant hypertension.
- **E — Eye Problems (Blurred Vision / Flashing Lights / Diplopia / Partial or Total Vision Loss):** Retinal artery or vein thrombosis, optic neuritis.
- **S — Severe Leg Pain / Calf Swelling (Unilateral warmth, erythema, tenderness):** Deep Vein Thrombosis (DVT) in the lower extremities.

5.3 Contraindications: WHO Medical Eligibility Criteria (MEC)

Trust Pills contain 30 mcg Ethinylestradiol. Estrogen stimulates hepatic synthesis of clotting factors (fibrinogen, factors VII, VIII, X) and downregulates antithrombin III, increasing thromboembolic risk.



WHO Category 4 Conditions (Absolute Contraindications)

1. **Smoking & Age ≥ 35:** Cigarette smoking ≥ 15 cigarettes/day in women aged 35 and older.
2. **Severe Hypertension:** Systolic Blood Pressure ≥ 160 mmHg or Diastolic Blood Pressure ≥ 100 mmHg, or hypertension with coexisting vascular disease.
3. **Active or Past Venous Thromboembolism:** History of or active Deep Vein Thrombosis (DVT) or Pulmonary Embolism (PE).
4. **Major Surgery with Prolonged Immobilization:** Elevated risk of postoperative venous stasis.
5. **Known Thrombophilic Mutations:** Factor V Leiden, Prothrombin G20210A mutation, Protein S, Protein C, or Antithrombin deficiencies.
6. **Ischemic Heart Disease & Stroke:** Current or past history of ischemic heart disease, myocardial infarction, or cerebrovascular accident.
7. **Complicated Valvular Heart Disease:** Pulmonary hypertension, risk of atrial fibrillation, history of subacute bacterial endocarditis.
8. **Migraine with Aura at ANY Age:** Significantly elevated risk of ischemic stroke. Symptoms include transient visual scotoma, scintillating lights, hemiparesthesia, or dysphasia preceding headaches.
9. **Current Breast Cancer:** Estrogen- and progesterone-dependent neoplasia.
10. **Severe Hepatic Disease:** Decompensated cirrhosis, active viral hepatitis flare, benign hepatic adenoma, or malignant hepatocellular carcinoma.
11. **Diabetes with Vascular Complications:** Diabetic nephropathy, retinopathy, neuropathy, or diabetes duration > 20 years.
12. **Postpartum (< 21 Days):** Elevated baseline postpartum hypercoagulability.

WHO Category 3 Conditions (Risks Generally Outweigh Advantages)

1. **Smoking & Age ≥ 35 (< 15 cigarettes/day):** Elevated cardiovascular risk profile.
2. **Moderately Controlled / Stage 1 Hypertension:** Systolic 140 to 159 mmHg or Diastolic 90 to 99 mmHg.
3. **Past Breast Cancer:** With no evidence of disease recurrence for at least 5 years.
4. **Migraine WITHOUT Aura in Women Aged ≥ 35:** Initiation of combined oral contraceptives is Category 3.
5. **Concurrent Enzyme-Inducing Drug Therapies:** Accelerates hepatic clearance of contraceptive steroids.

5.4 Pharmacokinetic Drug Interactions

Potent Hepatic Enzyme Inducers (CYP3A4 / P-Glycoprotein Inducers)

These medications induce cytochrome P450 3A4 (CYP3A4) in the liver and gut wall, accelerating the metabolic breakdown of ethinylestradiol and levonorgestrel. This lowers serum hormone concentrations below the ovulation-suppression threshold, causing method failure.

1. **Rifamycins:** Rifampicin, Rifabutin (antibiotics used for tuberculosis and atypical mycobacteria). *Potent CYP3A4 inducers that reduce circulating estrogen and progestin by over 50%.*
2. **Anticonvulsants / Antiepileptic Drugs (AEDs):** Carbamazepine, Phenytoin, Phenobarbital, Primidone, Oxcarbazepine, Topiramate, Felbamate.
3. **Herbal Remedies:** St. John's Wort (*Hypericum perforatum*).
4. **Antiretrovirals (Certain NNRTIs & Protease Inhibitors):** Efavirenz, Ritonavir-boosted regimens.

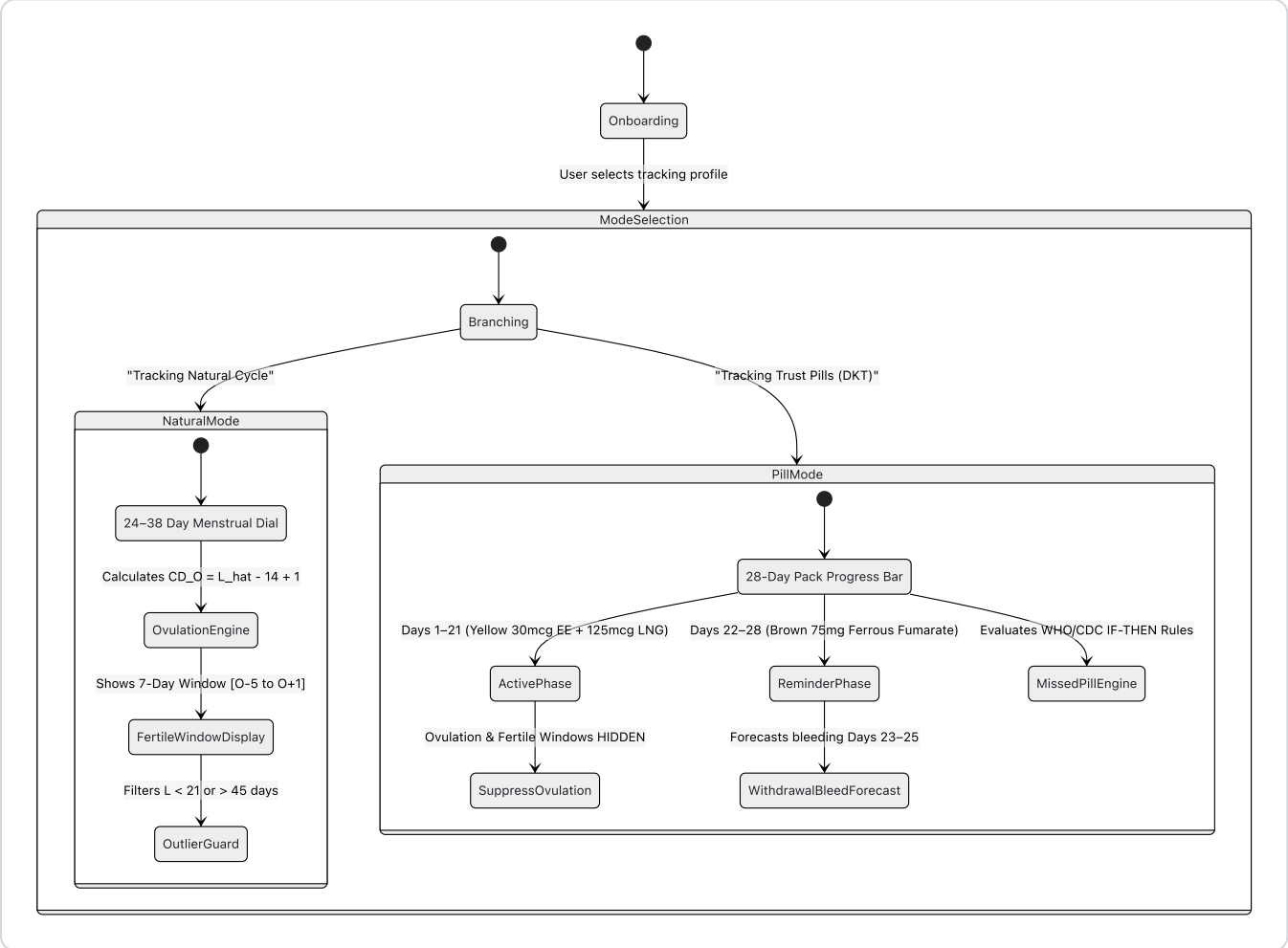
- **Clinical Rule for Petal App:** If a user logs any of these enzyme-inducing medications:
  - Recommend switching to a non-hormonal method (Copper IUD) or DMPA injections.
  - If continuing Trust Pills: **Mandate backup barrier methods (condoms) throughout co-administration AND for 28 consecutive days after discontinuing the enzyme inducer**.

The "Broad-Spectrum Antibiotic" Clarification (Non-Rifamycin)

- *Historical Myth:* Broad-spectrum antibiotics (such as amoxicillin, ampicillin, doxycycline, azithromycin, ciprofloxacin) were previously thought to lower COC efficacy by altering gut flora and reducing enterohepatic recycling of ethinylestradiol.
- *Modern Clinical Ground Truth (WHO & CDC Guidelines):* **Non-rifamycin broad-spectrum antibiotics DO NOT decrease the efficacy of combined oral contraceptives.** No additional barrier method is required when using these antibiotics unless severe vomiting or watery diarrhea occurs as a side effect.

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6. Petal App Architecture: System State Machine & Rule Matrix



6.1 State Machine Logic: Natural Cycle Mode vs. Trust Pill Mode

App Feature / Screen	Mode 1: Natural Menstrual Cycle	Mode 2: Trust Pills (DKT Philippines)
Primary Dashboard Dial	24–38 Day Cycle Counter (CD 1 to Next Period)	28-Day Blister Pack Counter (Yellow 1–21 → Brown 22–28)
Ovulation Calculation	Active: $\text{Next Period} - 14 \text{ days}$	**DISABLED / HIDDEN** (Ovulation pharmacologically suppressed)
Fertile Window Display	Active: 7-day probabilistic window ($O - 5$ to $O + 1$)	**DISABLED / HIDDEN** (No fertile window exists on COCs)
Bleeding Prediction	Next Natural Menses based on rolling average $\hat{L}$	Withdrawal Bleeding predicted on **Pack Days 23–25**
Daily Logging Prompts	Cervical Mucus, Basal Body Temperature, Libido, PMS	Pill Taken Time, Missed Pill Alarm, Spotting, Nausea, ACHES
Algorithm Engine	Moving Average & Outlier Detection	Deterministic Missed-Pill Decision Tree (WHO/CDC)
Emergency Guidance	Unprotected sex on fertile days → EC advice	Missed active pills → Specific pack-day recovery rules

6.2 Edge Case Logic Matrix for Software Engineers

Event / Trigger	Pre-Condition	App Decision & User Notification
User switches from Natural to Trust Pills	Cycle Day 1 of Period	Set Pack to Yellow Pill 1. Immediately hide fertile window and ovulation forecast. Notify: *"Trust Pills protect you immediately when started on Day 1. No backup needed."*
User switches from Natural to Trust Pills	Cycle Day 6+ (Outside menses)	Verify pregnancy is excluded. Set Pack to Yellow Pill 1. Display banner: *"Quick Start active: Use condoms for the next 7 consecutive days."*
Missed 1 Yellow Pill (<24h late)	Active Phase (Days 1–21)	Display: *"Take your late pill right away. Take today's pill at your usual time. You are still fully protected."*
Missed 2+ Yellow Pills in Week 1	Pack Days 1–7	Display: *"Take the most recent pill now. Discard other missed pills. Use condoms for 7 days. If you had sex in the past 5 days, emergency contraception is recommended."*
Missed 2+ Yellow Pills in Week 3	Pack Days 15–21	Display: *"Take most recent pill now. Finish the yellow pills. DISCARD ALL 7 BROWN PILLS. Start a new pack immediately tomorrow. Use condoms for 7 days."*
Missed Any Brown Pill	Pack Days 22–28	Display: *"Discard the missed brown pill. Brown pills contain iron, not hormones. Your protection is unaffected. Start your new pack on time."*
User logs "Severe calf pain" or "Chest pain"	Any Pack Day	**HIGH PRIORITY ALERT:** Display ACHES warning screen: *"Urgent: This may indicate a serious medical complication. Stop taking Trust Pills and seek emergency medical evaluation immediately."*